PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
digits = load_digits()
X_train, X_test, y_train, y_test = train_test_split(
    digits.data, digits.target, test_size=0.25, random_state=1)
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Accuracy:", round(accuracy_score(y_test, pred) * 100, 2), "%")
i = 0
print("\nTest image:")
for row in X_test[i].reshape(8, 8):
    print("".join("#" if p > 7 else "." for p in row))
print("Actual digit   :", y_test[i])
print("Predicted digit:", pred[i])
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ===============
Accuracy: 99.11 %

Test image:
...##...
..###...
..###...
..###...
..###...
..###...
..###...
...##...
Actual digit   : 1
Predicted digit: 1
>>>
```